

AI IN HEALTHCARE

AI-Driven Medical Diagnosis System

*Fast, explainable clinical decision support powered by
hybrid symbolic + probabilistic reasoning.*

Project Presentation · 2026

THE PROBLEM

Clinicians need faster, explainable decision support

Medical diagnosis is time-critical, knowledge-intensive, and error-prone. Clinicians juggle huge symptom spaces, overlapping conditions, and limited bedside time — while patients expect rapid, transparent answers.

- 1 Slow triage** Manual lookup across guidelines delays care.
- 2 Opaque AI tools** Black-box models that clinicians cannot trust.
- 3 Inconsistent reasoning** Outcomes vary by experience and workload.

WHY IT MATTERS

70%

of diagnostic errors are linked to cognitive overload and incomplete information gathering.

Source: clinical diagnostic-error literature

A hybrid reasoning engine, end to end

A desktop application that combines a structured medical knowledge base with classical AI search and probabilistic inference to produce diagnoses a clinician can read, question, and trust.

Symbolic

Structured knowledge base, symptom-driven rules, CSP constraints.

Search

A* search to navigate the diagnostic space efficiently.

Probabilistic

Bayesian updates that turn evidence into ranked likelihoods.

Hybrid

Symbolic + probabilistic outputs unified into one ranked report.

Layered modules, one diagnostic flow

Presentation

Polished Tkinter GUI: patient intake, symptom picker, results, charts, reports.

Reasoning

A* search + CSP for symptom matching, Bayesian inference for likelihoods, hybrid fusion.

Knowledge

Diseases, symptoms, prior probabilities, and constraints encoded as structured rules.

Storage

SQLite database for patients, sessions, symptoms, and persisted diagnostic reports.

DATA FLOW

- Patient input
- Symptoms
- A* + CSP
- Bayesian update
- Hybrid score
- Ranked diagnosis
- Saved report

How the diagnosis is actually computed

1 A* Search

Treats diagnosis as a search problem. A heuristic ranks candidate diseases by symptom overlap and prior likelihood, expanding only the most promising branches.

2 CSP

Symptoms, contraindications, and patient attributes act as constraints. The solver prunes diseases that violate hard rules (age, sex, exclusivity).

3 Bayesian Inference

Prior probabilities are updated with observed symptoms using likelihoods from the knowledge base to produce calibrated posterior scores.

4 Hybrid Fusion

Symbolic match strength and Bayesian posterior are combined into a single ranked list with a confidence value per diagnosis.

From intake to report in four steps

01

Patient Info

Capture demographics and history in a guided form.

02

Symptom Picker

Search, filter, and select symptoms from the knowledge base.

03

Automated Diagnosis

Hybrid engine runs A*, CSP, and Bayesian inference behind the scenes.

04

Results & Report

Ranked diagnoses, charts, and an exportable PDF-style report.

A polished interface, demo-ready

The interface was redesigned for clarity and demo presentation — a modern Tkinter shell, consistent visual language, and friendlier interaction patterns.



Modern theme

Cohesive color palette, typography, and spacing.



Guided forms

Step-by-step intake with validation and hints.



Smart symptom picker

Searchable list with categories and quick filters.



Result visualization

Bar/pie charts for ranked diagnoses and confidence.



One-click report

Exportable, printable diagnostic report.



Responsive layout

Resizable panels that fit projector and laptop.

What the system delivers

100%

Explainable

Every diagnosis is traced back to symptoms, rules, and probabilities.

<2s

Real-time

Hybrid inference returns ranked results in seconds, not minutes.

Demo-ready

Polished UI

Modernized Tkinter interface, charts, and exportable reports.

Interpretability

Clinicians see why, not just what.

Decision support

Augments — never replaces — judgment.

Persistence

Sessions and reports stored in SQLite.

Extensible

Knowledge base can grow with new diseases.

THANK YOU

Ready for demo.

AI-Driven Medical Diagnosis System · Hybrid reasoning for explainable, real-time clinical support.

Questions welcome.